
7 SEPTEMBER 2026 - 31 MARCH 2027

PRACTICAL AUTOMATION AND CONTROLS SYLLABUS

15 role-aligned modules with activities, exercises, case studies, projects, testing, and portfolio evidence

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Target: Junior / Trainee Automation, Controls, PLC, Instrumentation, and SCADA roles

17 hours per week | PC-first practical work | Evidence before claims

Role alignment and realistic target standard

Target role	Primary capability	Priority
Junior Automation / Controls	Electrical, PLC I/O and logic, HMI, motors/VFDs, networking, troubleshooting, documentation	Core
Junior PLC Technician	Scan cycle, I/O, Ladder Logic, modes, sequences, diagnostics, backups, and testing	Core
Junior Instrumentation	Measurement, 4-20 mA, scaling, valves, loops, P&IDs, and fault symptoms	Role-dependent
Junior SCADA Technician	Tags, HMI, alarms, trends, history concepts, communications, quality, and recovery	Core for SCADA
Automation Technician Trainee	Broad foundation, safe boundaries, practical evidence, integration, and learning under supervision	Core
Control Systems Technician	Requirements, PLC/HMI/SCADA, networks, testing, change control, and lifecycle documents	Core

Honest target

This workbook prepares you to contribute as a junior under supervision. It does not replace trade qualifications, site authorization, safety training, manufacturer training, or real plant experience.

EVIDENCE STANDARD USED THROUGHOUT

- | | |
|--|---|
| ☐ Studied - can explain the concept | ☐ Simulated - built in training software |
| ☐ Implemented - configured or programmed yourself | ☐ Tested - actual result recorded |
| ☐ Documented - usable technical record exists | ☐ Certified - valid certificate is held |

Program map: 15 practical modules

No.	Dates	Module	Practical deliverable
1	7-13 September 2026	Electrical quantities and safe study practice	Electrical calculation and circuit evidence pack
2	14-20 September 2026	Industrial components, control circuits, and schematics	24 VDC motor-control circuit simulation
3	21 September-4 October 2026	PLC hardware, scan cycle, addressing, and I/O	PLC I/O mapping and signal-trace mini-project
4	5-11 October 2026	Ladder Logic core patterns	PLC logic pattern workbook
5	12 October-1 November 2026	Modes, interlocks, permissives, sequences, and faults	Automated Conveyor System
6	2-15 November 2026	Motors, starters, protection, and VFD fundamentals	Motor and VFD interface mini-project
7	16-29 November 2026	Sensors and measurement principles	Instrument selection and measurement-chain study
8	30 November-13 December 2026	Analog signals, 4-20 mA, scaling, and calibration	4-20 mA loop and scaling laboratory
9	14 December 2026-3 January 2027	Process control, valves, alarms, and PID fundamentals	Automated Tank / Filling System
10	4-10 January 2027	HMI/SCADA design, alarms, trends, and operator usability	HMI operator and diagnostic screen pack
11	11-24 January 2027	Industrial networking and control-system communications	Industrial communications laboratory
12	25-31 January 2027	SCADA integration, diagnostics, and historical data	SCADA Process Control System
13	1-7 February 2027	Maintenance troubleshooting, testing, and change control	Fault-finding and test pack
14	8-28 February 2027	Flagship integrated automation system	Automated Water Treatment Plant - Personal Training / Simulation Project
15	1-31 March 2027	Portfolio, technical interviews, and targeted applications	Application readiness pack and technical demonstration

Three-page module pattern

Page 1 defines what to learn. Page 2 contains five activities, six exercises, and a realistic case study. Page 3 provides the dated Monday-to-Sunday plan, practical deliverable, evidence list, and verification record.

Master weekly learning cycle

Day	Time	Discipline	Required structure
Monday	2 h	Electrical	Recall 20 min learn 60 min calculate, draw, or trace 40 min
Tuesday	2 h	PLC	Recall 20 min learn 50 min program and test 50 min
Wednesday	2 h	Instrumentation	Recall 20 min learn 60 min calculate, draw, or simulate 40 min
Thursday	2 h	Process control	Recall 20 min learn 60 min industrial scenario 40 min
Friday	2 h	A-B-C-D rotation	Review 20 min rotating topic 60 min save evidence 40 min
Saturday	4 h	Practical project	Plan 60 min build 120 min test and document 60 min
Sunday	3 h	Recall and fault finding	Closed-book recall 60 min case/fault 60 min evidence review 60 min

EVERY WEEK MUST PRODUCE

- | | |
|--|---|
| ☐ One calculation, diagram, or electrical trace | ☐ One PLC routine or diagnostic task |
| ☐ One instrumentation application | ☐ One process-control explanation |
| ☐ One Friday rotation artifact | ☐ One tested project improvement |
| ☐ One completed fault or case study | ☐ One documentation or evidence update |

PC-first practical rule

Simulation is valid evidence when labelled honestly. Use existing low-voltage components only when you understand the circuit and can work safely. Never use mains voltage, defeat protection, or present simulation as a live industrial installation.

Friday rotation: September-December 2026

Friday	Rotation	Required focus/output
11 Sep	A - HMI/SCADA	Purpose, overview, and screen hierarchy
18 Sep	B - Networking	Ethernet and device addressing
25 Sep	C - Documentation	I/O list and tag conventions
2 Oct	D - Integration	Electrical-to-PLC signal path
9 Oct	A - HMI/SCADA	Controls, modes, states, and alarms
16 Oct	B - Networking	PLC/HMI communication path
23 Oct	C - Documentation	Conveyor narrative and FAT plan
30 Oct	D - Integration	Conveyor FAT and fault review
6 Nov	A - HMI/SCADA	Motor/VFD faceplate
13 Nov	B - Networking	Drive communication awareness
20 Nov	C - Documentation	Instrument list and signal chains
27 Nov	D - Integration	Motor/sensor project review
4 Dec	A - HMI/SCADA	Analog faceplate, alarms, and trends
11 Dec	B - Networking	Serial and HART fundamentals
18 Dec	C - Documentation	Tank P&ID, narrative, and alarms
25 Dec	D - Integration	Optional catch-up; no new material

Rotation key

A = HMI/SCADA. B = Industrial Networking. C = Engineering Documentation. D = Integration and Review. Keep the sequence when the month changes.

Friday rotation: January-March 2027

Friday	Rotation	Required focus/output
1 Jan	A - HMI/SCADA	Light tank-HMI review
8 Jan	B - Networking	IP plan and PLC/HMI path
15 Jan	C - Documentation	Network and protocol map
22 Jan	D - Integration	Communication recovery tests
29 Jan	A - HMI/SCADA	SCADA overview and diagnostics
5 Feb	B - Networking	Layered network fault finding
12 Feb	C - Documentation	Flagship baseline and traceability
19 Feb	D - Integration	Subsystem integration
26 Feb	A - HMI/SCADA	Flagship final HMI and trends
5 Mar	B - Networking	Interview-ready network explanation
12 Mar	C - Documentation	Portfolio evidence audit
19 Mar	D - Integration	Mock technical demonstration
26 Mar	A - HMI/SCADA	Final SCADA review

Rotation key

A = HMI/SCADA. B = Industrial Networking. C = Engineering Documentation. D = Integration and Review. Keep the sequence when the month changes.

MODULE 01

Electrical quantities and safe study practice

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	7-13 September 2026	ROLE ALIGNMENT	Essential for every target role
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Module objective

Build the electrical vocabulary, calculation ability, and safe study boundaries required before PLC I/O, instruments, motors, or control panels.

CORE CONTENT CHECKLIST

☐ Voltage, current, resistance, conductors, and insulators	☐ Ohm's Law and formula rearrangement
☐ Electrical power, watts, kilowatts, and energy	☐ AC versus DC, polarity, and frequency
☐ Series and parallel circuits	☐ Open circuit, short circuit, overload, and earth-fault concepts
☐ Electrical symbols and current-path tracing	☐ Isolation, LOTO awareness, and test-before-touch
☐ Multimeter functions and safe limits	☐ Knowledge versus supervision and formal authorization

REQUIRED LEARNING OUTCOMES

☐ Explain the module concepts without copying notes	☐ Complete the calculations, diagrams, or logic correctly
☐ Apply the content in the module practical activity	☐ Diagnose the module case study systematically
☐ Create the required engineering evidence	☐ State safety boundaries and assumptions honestly

COMPLETION GATE

☐ Closed-book explanation completed	☐ Exercises corrected and understood
☐ Practical behavior matches written expectation	☐ Fault root cause supported by evidence
☐ Documents are readable, dated, and versioned	☐ MODULE COMPLETE - date: _____

MODULE 01

Electrical quantities and safe study practice

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Create an electrical quantities map showing symbol, unit, instrument, and meaning	One-page concept artifact
2	Draw or calculate	Draw and label one series and one parallel circuit	Calculation or annotated diagram
3	Apply in simulation	Predict values, then verify them in a simulator	Screenshot plus configuration/logic
4	Test and observe	Create a safe-work boundary note for simulated, low-voltage, and supervised work	Expected-vs-actual result
5	Explain and document	Explain the four electrical quantities aloud without notes	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|--|--|
| ☐ A 24 VDC supply feeds 240 ohms: calculate current and power | ☐ A 12 V load draws 0.25 A: calculate resistance and power |
| ☐ Three 100 ohm resistors are in series on 24 VDC: calculate total resistance and current | ☐ Two 240 ohm resistors are parallel on 24 VDC: calculate equivalent resistance and current |
| ☐ Predict an open-circuit result and explain it | ☐ List the safe checks before measuring an unknown circuit |

CASE STUDY

A control-panel indicator is dark

SITUATION	A simulated 24 VDC panel has a healthy power supply, fuse, selector, and indicator. The supply is on, but the indicator is off.
AVAILABLE EVIDENCE	Supply output is 24 VDC. Voltage before the fuse is 24 VDC. Voltage after the fuse is 0 VDC. Indicator resistance is plausible.
YOUR TASKS	☐ Draw the expected current path ☐ Identify the most likely fault ☐ Name the next safe confirmation check ☐ State the correction and verification test
REQUIRED OUTPUT	One-page fault report with expected state, observations, root cause, correction, and verification.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 01

Electrical quantities and safe study practice

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
7-13 Sep	Mon - 2 h	Electrical quantities	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	PLC purpose preview	20 recall 50 learn 50 program/test
	Wed - 2 h	Digital/analog preview	20 recall 60 learn 40 apply
	Thu - 2 h	Variables and feedback preview	20 recall 60 learn 40 process case
	Fri - 2 h	11 Sep - A: HMI/SCADA overview	20 review 60 rotation 40 evidence
	Sat - 4 h	Build and test circuits	60 plan 120 build 60 test/document
	Sun - 3 h	Recall quiz; panel-lamp case; evidence	60 recall 60 fault 60 document/review

Project start

Saturday, 12 September 2026. Start after the relevant safety and readiness checks are complete.

Practical deliverable: Electrical calculation and circuit evidence pack

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|--|---|
| ☐ Calculation sheet | ☐ Series/parallel drawings |
| ☐ Simulator screenshots | ☐ Expected-vs-actual table |
| ☐ Fault report | ☐ Safety boundary note |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 02

Industrial components, control circuits, and schematics

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	14-20 September 2026	ROLE ALIGNMENT	Controls, PLC, automation, and instrumentation trainee roles
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Module objective

Recognize common components, trace command and feedback paths, and read a basic motor-control schematic.

CORE CONTENT CHECKLIST

☐ NO and NC contacts in the de-energized state	☐ Relay and contactor coils, poles, and auxiliary contacts
☐ Fuses, breakers, overload relays, and protection	☐ Control circuit versus power circuit
☐ Two-wire and three-wire control	☐ Seal-in or holding-contact operation
☐ Fail-safe stop and protective contacts	☐ Terminal blocks, wire numbers, tags, and cross-references
☐ Interface relays and PLC output loading	☐ E-stop and safety-relay awareness

REQUIRED LEARNING OUTCOMES

☐ Explain the module concepts without copying notes	☐ Complete the calculations, diagrams, or logic correctly
☐ Apply the content in the module practical activity	☐ Diagnose the module case study systematically
☐ Create the required engineering evidence	☐ State safety boundaries and assumptions honestly

COMPLETION GATE

☐ Closed-book explanation completed	☐ Exercises corrected and understood
☐ Practical behavior matches written expectation	☐ Fault root cause supported by evidence
☐ Documents are readable, dated, and versioned	☐ MODULE COMPLETE - date: _____

MODULE 02

Industrial components, control circuits, and schematics

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Label every component on a DOL start/stop schematic	One-page concept artifact
2	Draw or calculate	Color-code control and power paths	Calculation or annotated diagram
3	Apply in simulation	Create a state table for stop, start, run-latched, and overload states	Screenshot plus configuration/logic
4	Test and observe	Convert relay behavior into PLC inputs, logic, and outputs	Expected-vs-actual result
5	Explain and document	Mark five test points and expected states	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|--|--|
| ☐ Explain why a stop button is commonly NC | ☐ Explain why a PLC output normally does not drive a motor directly |
| ☐ Trace the action from Start press to latched run | ☐ Predict an open-overload-contact result |
| ☐ Draw Start, Stop, overload, coil, and holding contact | ☐ List three different causes of motor-will-not-start |

CASE STUDY	
Contactor energizes only while Start is held	
SITUATION	A simulated starter energizes when Start is pressed but drops out immediately when Start is released.
AVAILABLE EVIDENCE	Stop and overload contacts are healthy. The coil energizes while Start is true. The auxiliary holding contact never changes state.
YOUR TASKS	☐ Trace the seal-in path ☐ Classify the faulty layer ☐ State two possible root causes ☐ Design a test that distinguishes the causes
REQUIRED OUTPUT	Annotated schematic, state table, and troubleshooting report.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 02

Industrial components, control circuits, and schematics

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
14-20 Sep	Mon - 2 h	Relays/contactors/protection	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Map contacts to PLC concepts	20 recall 50 learn 50 program/test
	Wed - 2 h	Sensor switch states	20 recall 60 learn 40 apply
	Thu - 2 h	Permissive chain	20 recall 60 learn 40 process case
	Fri - 2 h	18 Sep - B: Ethernet basics	20 review 60 rotation 40 evidence
	Sat - 4 h	Simulate control circuit	60 plan 120 build 60 test/document
	Sun - 3 h	Seal-in fault; report	60 recall 60 fault 60 document/review

Project start

Saturday, 19 September 2026. Start after the relevant safety and readiness checks are complete.

Practical deliverable: 24 VDC motor-control circuit simulation

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- ☐ Motor-control schematic
- ☐ State table
- ☐ Fault screenshot
- ☐ Component list
- ☐ Test-point sheet
- ☐ Verification result

VERIFICATION RECORD

- ☐ Scope and assumptions written
- ☐ Normal behavior tested
- ☐ Actual results recorded honestly
- ☐ Files backed up and versioned
- ☐ Expected results written before testing
- ☐ At least two fault paths tested
- ☐ Unperformed tests marked NOT TESTED
- ☐ Evidence explainable without notes

MODULE 03

PLC hardware, scan cycle, addressing, and I/O

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	21 September-4 October 2026	ROLE ALIGNMENT	PLC, controls, SCADA, and automation technician roles
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Module objective

Understand where signals enter and leave a PLC, how the CPU executes logic, and how to diagnose a signal path.

CORE CONTENT CHECKLIST

☐ CPU, power supply, local/remote I/O, and communication modules	☐ Input scan, program execution, output update, and housekeeping
☐ Digital inputs and outputs	☐ Analog input and output purpose
☐ Physical addresses, tags, and internal memory	☐ Local versus remote I/O
☐ Module and channel diagnostics	☐ PNP/NPN and sourcing/sinking awareness
☐ Field state through HMI state	☐ Online monitoring, forcing risk, backups, and versions

REQUIRED LEARNING OUTCOMES

☐ Explain the module concepts without copying notes	☐ Complete the calculations, diagrams, or logic correctly
☐ Apply the content in the module practical activity	☐ Diagnose the module case study systematically
☐ Create the required engineering evidence	☐ State safety boundaries and assumptions honestly

COMPLETION GATE

☐ Closed-book explanation completed	☐ Exercises corrected and understood
☐ Practical behavior matches written expectation	☐ Fault root cause supported by evidence
☐ Documents are readable, dated, and versioned	☐ MODULE COMPLETE - date: _____

MODULE 03

PLC hardware, scan cycle, addressing, and I/O

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Draw a PLC architecture with CPU, I/O, network, HMI, and field devices	One-page concept artifact
2	Draw or calculate	Build a 12-point I/O list	Calculation or annotated diagram
3	Apply in simulation	Create a tag naming standard	Screenshot plus configuration/logic
4	Test and observe	Trace one input and one output through all layers	Expected-vs-actual result
5	Explain and document	Simulate a short pulse and relate it to scan time	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|---|--|
| ☐ Classify Start, contactor, transmitter, and valve-position command by I/O type | ☐ Explain the scan cycle in five sentences |
| ☐ List three causes for sensor active but PLC input false | ☐ List three causes for logic true but field output off |
| ☐ Explain forcing hazards | ☐ Create a reusable signal-trace checklist |

CASE STUDY

Proximity sensor LED is on but PLC input is off

SITUATION	An object is present and the sensor LED turns on. The HMI still shows Object Sensor Off and the conveyor sequence does not continue.
AVAILABLE EVIDENCE	Sensor supply is present. The input module has no channel LED. The PLC tag is false. Other channels operate normally.
YOUR TASKS	☐ Separate field, wiring, module, tag, and HMI possibilities ☐ List checks in safe order ☐ Select the likely fault ☐ Define the verification test
REQUIRED OUTPUT	Signal-path diagram and evidence-based fault report without random code changes.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 03

PLC hardware, scan cycle, addressing, and I/O

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
21-27 Sep	Mon - 2 h	24 VDC I/O paths	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	PLC hardware and scan	20 recall 50 learn 50 program/test
	Wed - 2 h	DI/DO devices	20 recall 60 learn 40 apply
	Thu - 2 h	Command-feedback path	20 recall 60 learn 40 process case
	Fri - 2 h	25 Sep - C: I/O list	20 review 60 rotation 40 evidence
	Sat - 4 h	Build I/O map	60 plan 120 build 60 test/document
	Sun - 3 h	Diagnose missing input	60 recall 60 fault 60 document/review
28 Sep-4 Oct	Mon - 2 h	PNP/NPN awareness	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Tags and diagnostics	20 recall 50 learn 50 program/test
	Wed - 2 h	AI/AO overview	20 recall 60 learn 40 apply
	Thu - 2 h	Timing effects	20 recall 60 learn 40 process case
	Fri - 2 h	2 Oct - D: integration	20 review 60 rotation 40 evidence
	Sat - 4 h	Complete signal tracing	60 plan 120 build 60 test/document
	Sun - 3 h	Diagnose missing output; sign off	60 recall 60 fault 60 document/review

Project start

Saturday, 26 September 2026. Start after the relevant safety and readiness checks are complete.

Practical deliverable: PLC I/O mapping and signal-trace mini-project

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- ☐ PLC architecture
- ☐ Tag standard
- ☐ Input fault trace
- ☐ Backup record
- ☐ I/O list
- ☐ Monitor screenshots
- ☐ Output fault trace

VERIFICATION RECORD

- ☐ Scope and assumptions written
- ☐ Normal behavior tested
- ☐ Actual results recorded honestly
- ☐ Files backed up and versioned
- ☐ Expected results written before testing
- ☐ At least two fault paths tested
- ☐ Unperformed tests marked NOT TESTED
- ☐ Evidence explainable without notes

MODULE 04

Ladder Logic core patterns

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	5-11 October 2026	ROLE ALIGNMENT	Essential PLC and controls programming foundation
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Module objective

Read, write, comment, and test simple Ladder Logic before building modes or automated sequences.

CORE CONTENT CHECKLIST

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|---|---|
| ☐ Rung evaluation order | ☐ Examine-on and examine-off conditions |
| ☐ Output coils and internal bits | ☐ Start/stop, seal-in, latch, and unlatch |
| ☐ On-delay and off-delay timers | ☐ Counters, reset, and one-shots |
| ☐ Comparison instructions | ☐ Retentive and non-retentive behavior |
| ☐ Duplicate-coil and reset risks | ☐ Comments, tag names, monitoring, and written tests |

REQUIRED LEARNING OUTCOMES

- | | |
|---|--|
| ☐ Explain the module concepts without copying notes | ☐ Complete the calculations, diagrams, or logic correctly |
| ☐ Apply the content in the module practical activity | ☐ Diagnose the module case study systematically |
| ☐ Create the required engineering evidence | ☐ State safety boundaries and assumptions honestly |

COMPLETION GATE

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|---|---|
| ☐ Closed-book explanation completed | ☐ Exercises corrected and understood |
| ☐ Practical behavior matches written expectation | ☐ Fault root cause supported by evidence |
| ☐ Documents are readable, dated, and versioned | ☐ MODULE COMPLETE - date: _____ |

MODULE 04

Ladder Logic core patterns

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Write and explain a start/stop rung from memory	One-page concept artifact
2	Draw or calculate	Build a five-second delayed output	Calculation or annotated diagram
3	Apply in simulation	Build an object counter with one-shot and reset	Screenshot plus configuration/logic
4	Test and observe	Create high alarm with a separate reset threshold	Expected-vs-actual result
5	Explain and document	Write truth tables before testing every routine	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|---|---|
| ☐ Truth table for series and parallel contacts | ☐ Compare seal-in with Set/Reset latch |
| ☐ Predict timer behavior when enable flickers | ☐ Explain why a one-shot is needed for a maintained sensor |
| ☐ Find a duplicate-coil error | ☐ Write five start/stop tests including power-up |

CASE STUDY

Batch counter increases several times for one object

SITUATION	A photoelectric sensor remains on for several scans while a box passes, and the batch target is reached too early.
AVAILABLE EVIDENCE	The sensor is true for 180 ms. Scan time is about 10 ms. The counter is enabled directly by the input.
YOUR TASKS	☐ Explain the scan-related cause ☐ Choose an edge or one-shot pattern ☐ Predict the corrected count for five objects ☐ Create a sensor-chatter test
REQUIRED OUTPUT	Corrected rung, timing sketch, and counter test table.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 04

Ladder Logic core patterns

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
5-11 Oct	Mon - 2 h	Control-circuit comparison	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Ladder patterns	20 recall 50 learn 50 program/test
	Wed - 2 h	Sensor input states	20 recall 60 learn 40 apply
	Thu - 2 h	Timing and state behavior	20 recall 60 learn 40 process case
	Fri - 2 h	9 Oct - A: HMI controls	20 review 60 rotation 40 evidence
	Sat - 4 h	Build logic workbook	60 plan 120 build 60 test/document
	Sun - 3 h	Counter case; logic quiz	60 recall 60 fault 60 document/review

Project start

Saturday, 10 October 2026. Start after the relevant safety and readiness checks are complete.

Practical deliverable: PLC logic pattern workbook

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|--|--|
| ☐ Commented ladder routines | ☐ Truth tables |
| ☐ Tag list | ☐ Timer/counter tests |
| ☐ Fault note | ☐ Revision log |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 05

Modes, interlocks, permissives, sequences, and faults

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	12 October-1 November 2026	ROLE ALIGNMENT	Automation, PLC, and controls technician roles
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Module objective

Design predictable behavior using command ownership, operating conditions, ordered steps, trips, alarms, reset rules, and safe recovery.

CORE CONTENT CHECKLIST

☐ Command, status, feedback, permissive, interlock, alarm, and trip	☐ Auto, Manual, Local, and Remote ownership
☐ Mode-transition behavior	☐ Sequence steps and transitions
☐ State-machine awareness and reset	☐ Start inhibit and restart policy
☐ Command-feedback discrepancy timer	☐ Fault latch, acknowledgement, correction, and reset
☐ First-out fault	☐ Narrative, sequence chart, and cause-and-effect

REQUIRED LEARNING OUTCOMES

☐ Explain the module concepts without copying notes	☐ Complete the calculations, diagrams, or logic correctly
☐ Apply the content in the module practical activity	☐ Diagnose the module case study systematically
☐ Create the required engineering evidence	☐ State safety boundaries and assumptions honestly

COMPLETION GATE

☐ Closed-book explanation completed	☐ Exercises corrected and understood
☐ Practical behavior matches written expectation	☐ Fault root cause supported by evidence
☐ Documents are readable, dated, and versioned	☐ MODULE COMPLETE - date: _____

MODULE 05

Modes, interlocks, permissives, sequences, and faults

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Create a conveyor permissive summary	One-page concept artifact
2	Draw or calculate	Define command ownership for every mode	Calculation or annotated diagram
3	Apply in simulation	Build a three-step sequence and transition diagram	Screenshot plus configuration/logic
4	Test and observe	Add motor feedback and start-failure timing	Expected-vs-actual result
5	Explain and document	Inject five faults and prove reset requires correction	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|--|---|
| ☐ Classify low pressure as permissive, alarm, or trip | ☐ Write a Boolean start-permission expression |
| ☐ Design Auto-to-Manual transition behavior | ☐ Explain acknowledgement versus reset |
| ☐ Write Idle-Starting-Running-Stopping transitions | ☐ Create cause-and-effect rows for four faults |

CASE STUDY

Conveyor sequence stalls after detecting a box

SITUATION	The conveyor detects a box and stops at transfer. The downstream conveyor never starts and the sequence remains in Step 3.
AVAILABLE EVIDENCE	Step 3 is active. Safety and overload are healthy. Downstream Ready is false. No trip alarm is active.
YOUR TASKS	☐ Explain why code changes are not the first action ☐ Trace Downstream Ready ☐ Define useful HMI diagnostics ☐ Write recovery and regression tests
REQUIRED OUTPUT	Sequence-state report, permissive trace, correction, and regression test.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 05

Modes, interlocks, permissives, sequences, and faults

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
12-18 Oct	Mon - 2 h	Motor-control review	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Permissives/interlocks	20 recall 50 learn 50 program/test
	Wed - 2 h	Object sensor	20 recall 60 learn 40 apply
	Thu - 2 h	Modes and ownership	20 recall 60 learn 40 process case
	Fri - 2 h	16 Oct - B: PLC/HMI comms	20 review 60 rotation 40 evidence
	Sat - 4 h	Build start/stop and modes	60 plan 120 build 60 test/document
	Sun - 3 h	Will-not-start drill	60 recall 60 fault 60 document/review
19-25 Oct	Mon - 2 h	Overload protection	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Sequence and counting	20 recall 50 learn 50 program/test
	Wed - 2 h	Sensor fault	20 recall 60 learn 40 apply
	Thu - 2 h	Command-feedback	20 recall 60 learn 40 process case
	Fri - 2 h	23 Oct - C: narrative/test	20 review 60 rotation 40 evidence
	Sat - 4 h	Add faults and HMI	60 plan 120 build 60 test/document
	Sun - 3 h	Sequence-stall case	60 recall 60 fault 60 document/review
26 Oct-1 Nov	Mon - 2 h	Isolation review	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Refactor/comments	20 recall 50 learn 50 program/test
	Wed - 2 h	Signal plausibility	20 recall 60 learn 40 apply
	Thu - 2 h	Safe recovery	20 recall 60 learn 40 process case
	Fri - 2 h	30 Oct - D: FAT review	20 review 60 rotation 40 evidence
	Sat - 4 h	Execute full tests	60 plan 120 build 60 test/document
	Sun - 3 h	Retest, document, sign off	60 recall 60 fault 60 document/review

Project start

Saturday, 17 October 2026. Start after the relevant safety and readiness checks are complete.

Practical deliverable: Automated Conveyor System

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|--|---|
| ☐ Process diagram | ☐ I/O list |
| ☐ Control narrative | ☐ PLC logic |
| ☐ HMI | ☐ Cause-and-effect |
| ☐ FAT results | ☐ Fault reports |
| ☐ Demo | |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 06

Motors, starters, protection, and VFD fundamentals

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	2-15 November 2026	ROLE ALIGNMENT	Controls, automation, PLC, and maintenance-oriented roles
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Module objective

Understand motor-control interfaces and diagnose command, starter, drive, feedback, and process conditions within safe boundaries.

CORE CONTENT CHECKLIST

☐ Three-phase motor fundamentals	☐ Motor nameplate voltage, current, frequency, power, and speed
☐ DOL and reversing starters	☐ Electrical and mechanical interlocking
☐ Overload and auxiliary feedback	☐ VFD purpose and power-conversion concept
☐ Run, direction, speed, ready, running, and fault signals	☐ Digital, analog, and network interfaces
☐ Acceleration, deceleration, and reset awareness	☐ Isolation and stored-energy awareness

REQUIRED LEARNING OUTCOMES

☐ Explain the module concepts without copying notes	☐ Complete the calculations, diagrams, or logic correctly
☐ Apply the content in the module practical activity	☐ Diagnose the module case study systematically
☐ Create the required engineering evidence	☐ State safety boundaries and assumptions honestly

COMPLETION GATE

☐ Closed-book explanation completed	☐ Exercises corrected and understood
☐ Practical behavior matches written expectation	☐ Fault root cause supported by evidence
☐ Documents are readable, dated, and versioned	☐ MODULE COMPLETE - date: _____

MODULE 06

Motors, starters, protection, and VFD fundamentals

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Trace a DOL starter from PLC to motor feedback	One-page concept artifact
2	Draw or calculate	Read a sample nameplate	Calculation or annotated diagram
3	Apply in simulation	Create a hardwired PLC-to-VFD signal map	Screenshot plus configuration/logic
4	Test and observe	Build a simulated motor faceplate	Expected-vs-actual result
5	Explain and document	Create a decision tree separating PLC, drive, motor, and process faults	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|--|--|
| ☐ Explain command versus running feedback | ☐ State overload-trip behavior |
| ☐ Calculate percentage speed for 35 Hz of a 50 Hz maximum | ☐ List four reasons a healthy VFD may not run |
| ☐ Explain acceleration and deceleration ramps | ☐ Identify checks requiring qualified personnel |

CASE STUDY

VFD receives a run command but the motor does not turn

SITUATION	The PLC command is true and speed reference is 60 percent. The drive display shows LOCAL and Ready is false.
AVAILABLE EVIDENCE	Communication is healthy. The drive control source is the keypad. No PLC output fault exists.
YOUR TASKS	☐ Identify the controlling layer ☐ Explain why PLC changes are inappropriate ☐ State the simulated correction ☐ Verify start, stop, speed, and fault response
REQUIRED OUTPUT	Command-source analysis, VFD signal map, and verified test record.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 06

Motors, starters, protection, and VFD fundamentals

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
2-8 Nov	Mon - 2 h	Motors and starters	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Motor interface logic	20 recall 50 learn 50 program/test
	Wed - 2 h	Motor feedback	20 recall 60 learn 40 apply
	Thu - 2 h	Command-feedback	20 recall 60 learn 40 process case
	Fri - 2 h	6 Nov - A: motor faceplate	20 review 60 rotation 40 evidence
	Sat - 4 h	Build starter simulation	60 plan 120 build 60 test/document
	Sun - 3 h	Overload diagnosis	60 recall 60 fault 60 document/review
9-15 Nov	Mon - 2 h	VFD fundamentals	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Speed/reset logic	20 recall 50 learn 50 program/test
	Wed - 2 h	Analog speed reference	20 recall 60 learn 40 apply
	Thu - 2 h	Drive status/trips	20 recall 60 learn 40 process case
	Fri - 2 h	13 Nov - B: drive comms	20 review 60 rotation 40 evidence
	Sat - 4 h	Complete VFD interface	60 plan 120 build 60 test/document
	Sun - 3 h	LOCAL/Ready case; sign off	60 recall 60 fault 60 document/review

Project start

Saturday, 7 November 2026. Start after the relevant safety and readiness checks are complete.

Practical deliverable: Motor and VFD interface mini-project

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|--|--|
| ☐ Starter schematic | ☐ Nameplate worksheet |
| ☐ VFD signal map | ☐ PLC logic |
| ☐ HMI faceplate | ☐ Test results |
| ☐ Fault report | |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 07

Sensors and measurement principles

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	16-29 November 2026	ROLE ALIGNMENT	Instrumentation, controls, PLC, SCADA, and automation roles
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Module objective

Understand how physical conditions become trustworthy values and how selection, installation, and process effects influence measurement.

CORE CONTENT CHECKLIST

- | | |
|--|--|
| ☐ Process to sensor to transmitter to PLC to HMI | ☐ Sensor versus transmitter |
| ☐ RTD, thermocouple, and temperature transmitter | ☐ Gauge, absolute, and differential pressure |
| ☐ Float, hydrostatic, ultrasonic, and radar level | ☐ Magnetic, ultrasonic, turbine, and DP flow awareness |
| ☐ LRV, URV, range, and span | ☐ Accuracy, repeatability, resolution, hysteresis, and response |
| ☐ Installation and process effects | ☐ Open, short, frozen, noisy, drifted, and implausible failures |

REQUIRED LEARNING OUTCOMES

- | | |
|---|--|
| ☐ Explain the module concepts without copying notes | ☐ Complete the calculations, diagrams, or logic correctly |
| ☐ Apply the content in the module practical activity | ☐ Diagnose the module case study systematically |
| ☐ Create the required engineering evidence | ☐ State safety boundaries and assumptions honestly |

COMPLETION GATE

- | | |
|---|---|
| ☐ Closed-book explanation completed | ☐ Exercises corrected and understood |
| ☐ Practical behavior matches written expectation | ☐ Fault root cause supported by evidence |
| ☐ Documents are readable, dated, and versioned | ☐ MODULE COMPLETE - date: _____ |

MODULE 07

Sensors and measurement principles

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Draw four measurement chains	One-page concept artifact
2	Draw or calculate	Compare three technologies for one service	Calculation or annotated diagram
3	Apply in simulation	Build an instrument list with tag, range, units, signal, and failure state	Screenshot plus configuration/logic
4	Test and observe	Create four range-and-span examples	Expected-vs-actual result
5	Explain and document	Classify six simulated trends by fault symptom	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|--|---|
| ☐ Select level technology for clean water | ☐ Select level technology for a foaming vessel |
| ☐ Calculate span for -1 to 9 bar | ☐ Explain accurate instrument but poor measurement |
| ☐ List ultrasonic installation problems | ☐ Write checks for a frozen temperature value |

CASE STUDY		Ultrasonic level jumps during tank filling
SITUATION	Tank level becomes erratic only while the inlet valve is open. It stabilizes when filling stops.	
AVAILABLE EVIDENCE	Communication and quality are healthy. Foam and turbulence form beneath the sensor during filling.	
YOUR TASKS	☐ Separate electrical, configuration, installation, and process causes ☐ Select the likely cause ☐ Suggest mitigations ☐ Define evidence that proves improvement	
REQUIRED OUTPUT	Instrument-selection review and trend-based troubleshooting report.	

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 07

Sensors and measurement principles

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
16-22 Nov	Mon - 2 h	Sensor power/wiring	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Input diagnostics	20 recall 50 learn 50 program/test
	Wed - 2 h	Temperature/pressure	20 recall 60 learn 40 apply
	Thu - 2 h	Measurement quality	20 recall 60 learn 40 process case
	Fri - 2 h	20 Nov - C: instrument list	20 review 60 rotation 40 evidence
	Sat - 4 h	Select and document	60 plan 120 build 60 test/document
	Sun - 3 h	Frozen-signal drill	60 recall 60 fault 60 document/review
23-29 Nov	Mon - 2 h	Shielding awareness	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Alarm from measurement	20 recall 50 learn 50 program/test
	Wed - 2 h	Level/flow	20 recall 60 learn 40 apply
	Thu - 2 h	Process effects	20 recall 60 learn 40 process case
	Fri - 2 h	27 Nov - D: integration	20 review 60 rotation 40 evidence
	Sat - 4 h	Complete measurement chain	60 plan 120 build 60 test/document
	Sun - 3 h	Ultrasonic case; sign off	60 recall 60 fault 60 document/review

Project start

Saturday, 21 November 2026. Start after the relevant safety and readiness checks are complete.

Practical deliverable: Instrument selection and measurement-chain study

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|---|---|
| ☐ Process sketch | ☐ Instrument list |
| ☐ Selection matrix | ☐ Range calculations |
| ☐ Signal chains | ☐ Trend analysis |
| ☐ Fault reports | |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 08

Analog signals, 4-20 mA, scaling, and calibration

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	30 November-13 December 2026	ROLE ALIGNMENT	Instrumentation, PLC, SCADA, controls, and automation roles
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Module objective

Connect field signals to engineering units, detect abnormal values, and troubleshoot analog loops systematically.

CORE CONTENT CHECKLIST

☐ Digital versus analog signals	☐ 4-20 mA live zero
☐ Two-wire, three-wire, and four-wire transmitters	☐ Loop supply and burden awareness
☐ Raw counts, current, percent, and engineering units	☐ Under-range, over-range, open-loop, and bad quality
☐ Noise, shielding, grounding, routing, and filtering	☐ Zero, span, LRV, URV, and trim
☐ As-found and as-left calibration records	☐ HART purpose and communicator awareness

REQUIRED LEARNING OUTCOMES

☐ Explain the module concepts without copying notes	☐ Complete the calculations, diagrams, or logic correctly
☐ Apply the content in the module practical activity	☐ Diagnose the module case study systematically
☐ Create the required engineering evidence	☐ State safety boundaries and assumptions honestly

COMPLETION GATE

☐ Closed-book explanation completed	☐ Exercises corrected and understood
☐ Practical behavior matches written expectation	☐ Fault root cause supported by evidence
☐ Documents are readable, dated, and versioned	☐ MODULE COMPLETE - date: _____

MODULE 08

Analog signals, 4-20 mA, scaling, and calibration

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Create 0-25-50-75-100 percent current tables for two ranges	One-page concept artifact
2	Draw or calculate	Build a two-direction scaling worksheet	Calculation or annotated diagram
3	Apply in simulation	Simulate two transmitters and compare expected with PLC values	Screenshot plus configuration/logic
4	Test and observe	Add abnormal-signal and plausibility alarms	Expected-vs-actual result
5	Explain and document	Draw a loop and mark supervised test points	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|---|--|
| ☐ Convert 12 mA for a 0-10 bar range | ☐ Convert 650 L/min for a 0-1000 L/min range to current |
| ☐ Convert 16 mA for a 2-8 m range | ☐ Explain 0 mA versus valid 0 percent |
| ☐ List causes of noisy analog input | ☐ Create an as-found/as-left template |

CASE STUDY	
Pressure display is consistently too high	
SITUATION	A new 0-10 bar transmitter produces about 12 mA at a trusted 5 bar condition, but the HMI displays 8 bar.
AVAILABLE EVIDENCE	Transmitter range is 0-10 bar. PLC scaling is configured for 0-16 bar. Signal and wiring are stable.
YOUR TASKS	☐ Calculate the correct value ☐ Identify the root cause ☐ Explain why transmitter calibration is wrong ☐ Correct scaling and test 0, 50, and 100 percent
REQUIRED OUTPUT	Scaling worksheet, corrected configuration, and three-point verification record.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 08

Analog signals, 4-20 mA, scaling, and calibration

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
30 Nov-6 Dec	Mon - 2 h	Analog wiring	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Raw values/scaling	20 recall 50 learn 50 program/test
	Wed - 2 h	4-20 mA loop	20 recall 60 learn 40 apply
	Thu - 2 h	PV quality	20 recall 60 learn 40 process case
	Fri - 2 h	4 Dec - A: analog HMI	20 review 60 rotation 40 evidence
	Sat - 4 h	Build first loop	60 plan 120 build 60 test/document
	Sun - 3 h	Open-loop case	60 recall 60 fault 60 document/review
7-13 Dec	Mon - 2 h	Noise/shielding	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Plausibility/alarms	20 recall 50 learn 50 program/test
	Wed - 2 h	Calibration/HART	20 recall 60 learn 40 apply
	Thu - 2 h	Filtering	20 recall 60 learn 40 process case
	Fri - 2 h	11 Dec - B: serial/HART	20 review 60 rotation 40 evidence
	Sat - 4 h	Test second loop	60 plan 120 build 60 test/document
	Sun - 3 h	Scaling case; sign off	60 recall 60 fault 60 document/review

Project start

Saturday, 5 December 2026. Start after the relevant safety and readiness checks are complete.

Practical deliverable: 4-20 mA loop and scaling laboratory

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|--|---|
| ☐ Loop diagram | ☐ Scaling calculations |
| ☐ PLC configuration | ☐ HMI/trend |
| ☐ Alarm tests | ☐ Calibration template |
| ☐ Fault report | |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 09

Process control, valves, alarms, and PID fundamentals

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	14 December 2026-3 January 2027	ROLE ALIGNMENT	Process automation, instrumentation, SCADA, and controls roles
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Module objective

Understand how measurements, setpoints, controllers, actuators, and processes interact so faults can be interpreted using process behavior.

CORE CONTENT CHECKLIST

☐ Process objective, inputs, outputs, and disturbances	☐ PV, SP, MV/CV, and feedback
☐ Open-loop and closed-loop control	☐ Manual, Automatic, and bumpless-transfer awareness
☐ Direct and reverse action	☐ Proportional, integral, and derivative effects
☐ Dead time, lag, overshoot, saturation, and hunting	☐ Valve, actuator, I/P, positioner, and feedback
☐ Air-to-open, air-to-close, and fail position	☐ Alarm deadband, priority, high-high trip, and recovery

REQUIRED LEARNING OUTCOMES

☐ Explain the module concepts without copying notes	☐ Complete the calculations, diagrams, or logic correctly
☐ Apply the content in the module practical activity	☐ Diagnose the module case study systematically
☐ Create the required engineering evidence	☐ State safety boundaries and assumptions honestly

COMPLETION GATE

☐ Closed-book explanation completed	☐ Exercises corrected and understood
☐ Practical behavior matches written expectation	☐ Fault root cause supported by evidence
☐ Documents are readable, dated, and versioned	☐ MODULE COMPLETE - date: _____

MODULE 09

Process control, valves, alarms, and PID fundamentals

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Draw and label three feedback loops	One-page concept artifact
2	Draw or calculate	Analyze stable, overshoot, oscillation, and saturation trends	Calculation or annotated diagram
3	Apply in simulation	Create an alarm list with operator response	Screenshot plus configuration/logic
4	Test and observe	Build Manual and Automatic level-control behavior	Expected-vs-actual result
5	Explain and document	Add and test a valve-position discrepancy alarm	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|--|--|
| ☐ Identify PV, SP, and MV in a level loop | ☐ Explain higher proportional action |
| ☐ Explain integral action and windup | ☐ Choose a cooling-valve fail position |
| ☐ Compare High alarm and High-High trip | ☐ Write controlled restart after a high-high trip |

CASE STUDY

Tank reaches high-high level while outlet command is open

SITUATION	Controller output and outlet-valve command are high, but tank level continues rising until high-high shutdown.
AVAILABLE EVIDENCE	Valve command is 100 percent. Position feedback remains 5 percent. Inlet flow continues. Communication is healthy.
YOUR TASKS	☐ Identify the failed loop element ☐ State the protective response ☐ Select trend signals ☐ Write recovery and verification steps
REQUIRED OUTPUT	Loop diagram, event timeline, cause-and-effect row, and verified fault report.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 09

Process control, valves, alarms, and PID fundamentals

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
14-20 Dec	Mon - 2 h	Actuator interfaces	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Tank sequence/modes	20 recall 50 learn 50 program/test
	Wed - 2 h	Valves/feedback	20 recall 60 learn 40 apply
	Thu - 2 h	PV/SP/feedback	20 recall 60 learn 40 process case
	Fri - 2 h	18 Dec - C: P&ID;	20 review 60 rotation 40 evidence
	Sat - 4 h	Build tank process	60 plan 120 build 60 test/document
	Sun - 3 h	Valve-will-not-open	60 recall 60 fault 60 document/review
21-27 Dec	Mon - 2 h	Protection review	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Alarm/trip logic	20 recall 50 learn 50 program/test
	Wed - 2 h	Instrument faults	20 recall 60 learn 40 apply
	Thu - 2 h	PID/trends	20 recall 60 learn 40 process case
	Fri - 2 h	25 Dec - D: catch-up	20 review 60 rotation 40 evidence
	Sat - 4 h	Add alarms/control	60 plan 120 build 60 test/document
	Sun - 3 h	High-high case	60 recall 60 fault 60 document/review
28 Dec-3 Jan	Mon - 2 h	Schematic review	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Refactor/comments	20 recall 50 learn 50 program/test
	Wed - 2 h	Range/plausibility	20 recall 60 learn 40 apply
	Thu - 2 h	Manual/Auto recovery	20 recall 60 learn 40 process case
	Fri - 2 h	1 Jan - A: light HMI	20 review 60 rotation 40 evidence
	Sat - 4 h	Execute tests	60 plan 120 build 60 test/document
	Sun - 3 h	Close documents; sign off	60 recall 60 fault 60 document/review

Project start

Saturday, 19 December 2026. Start after the relevant safety and readiness checks are complete.

Practical deliverable: Automated Tank / Filling System

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|--|---|
| ☐ P&ID; | ☐ I/O and instrument lists |
| ☐ Control narrative | ☐ PLC logic |
| ☐ HMI/trends | ☐ Alarm list |
| ☐ Test procedure | ☐ Fault reports |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 10

HMI/SCADA design, alarms, trends, and operator usability

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	4-10 January 2027	ROLE ALIGNMENT	SCADA, automation, controls, and PLC technician roles
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Module objective

Present process state clearly and avoid displays that hide mode, interlock, bad-quality, or communication information.

CORE CONTENT CHECKLIST

- | | |
|---|--|
| ☐ HMI versus SCADA purpose | ☐ Overview, detail, alarm, trend, and diagnostic displays |
| ☐ Operator versus maintenance information | ☐ Navigation hierarchy |
| ☐ Equipment faceplates and ownership | ☐ Consistent states, units, labels, and restrained color |
| ☐ Alarm priority, acknowledgement, and history | ☐ Trend signals, ranges, units, and comparison |
| ☐ Tag quality, timestamp, and stale data | ☐ User access, confirmation, and change awareness |

REQUIRED LEARNING OUTCOMES

- | | |
|---|--|
| ☐ Explain the module concepts without copying notes | ☐ Complete the calculations, diagrams, or logic correctly |
| ☐ Apply the content in the module practical activity | ☐ Diagnose the module case study systematically |
| ☐ Create the required engineering evidence | ☐ State safety boundaries and assumptions honestly |

COMPLETION GATE

- | | |
|---|---|
| ☐ Closed-book explanation completed | ☐ Exercises corrected and understood |
| ☐ Practical behavior matches written expectation | ☐ Fault root cause supported by evidence |
| ☐ Documents are readable, dated, and versioned | ☐ MODULE COMPLETE - date: _____ |

MODULE 10

HMI/SCADA design, alarms, trends, and operator usability

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Sketch a five-screen hierarchy	One-page concept artifact
2	Draw or calculate	Build an uncluttered process overview	Calculation or annotated diagram
3	Apply in simulation	Build a pump or valve faceplate	Screenshot plus configuration/logic
4	Test and observe	Configure five alarms with operator response	Expected-vs-actual result
5	Explain and document	Create a diagnostic trend with command, feedback, PV, SP, and quality	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|---|---|
| ☐ Choose overview versus maintenance information | ☐ Explain restrained alarm color use |
| ☐ Specify display behavior during communication loss | ☐ Design high-impact command confirmation |
| ☐ Select trend signals for pump diagnosis | ☐ Perform a 30-second recruiter usability review |

CASE STUDY

Operator sees a normal level after PLC communication fails

SITUATION	The network is removed in simulation. Level remains at 62 percent with no visible bad-quality indication.
AVAILABLE EVIDENCE	The last value remains displayed. Timestamp stops changing. Communication status becomes false but is not shown.
YOUR TASKS	☐ Identify the HMI design failure ☐ Define stale-data appearance ☐ Add status and alarm behavior ☐ Test loss and recovery
REQUIRED OUTPUT	Before/after screen evidence, alarm test, and usability explanation.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 10

HMI/SCADA design, alarms, trends, and operator usability

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
4-10 Jan	Mon - 2 h	Panel/HMI power	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	HMI tag interface	20 recall 50 learn 50 program/test
	Wed - 2 h	PV quality display	20 recall 60 learn 40 apply
	Thu - 2 h	Operator controls	20 recall 60 learn 40 process case
	Fri - 2 h	8 Jan - B: IP/tag path	20 review 60 rotation 40 evidence
	Sat - 4 h	Build screen pack	60 plan 120 build 60 test/document
	Sun - 3 h	Stale-data case	60 recall 60 fault 60 document/review

Project start
Saturday, 9 January 2027. Start after the relevant safety and readiness checks are complete.

Practical deliverable: HMI operator and diagnostic screen pack

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- ☐ Screen hierarchy
- ☐ Equipment faceplate
- ☐ Trend analysis
- ☐ Usability checklist
- ☐ Overview
- ☐ Alarm tests
- ☐ Comms-loss display

VERIFICATION RECORD

- ☐ Scope and assumptions written
- ☐ Normal behavior tested
- ☐ Actual results recorded honestly
- ☐ Files backed up and versioned
- ☐ Expected results written before testing
- ☐ At least two fault paths tested
- ☐ Unperformed tests marked NOT TESTED
- ☐ Evidence explainable without notes

MODULE 11

Industrial networking and control-system communications

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	11-24 January 2027	ROLE ALIGNMENT	SCADA, controls, automation, and PLC technician roles
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Module objective

Understand industrial data paths and diagnose physical, link, addressing, protocol, mapping, and application problems in order.

CORE CONTENT CHECKLIST

- | | |
|--|---|
| ☐ Ethernet media, links, switches, and topology | ☐ MAC and IPv4 addresses |
| ☐ Subnet masks, gateways, and routing awareness | ☐ Ping and connectivity tests |
| ☐ TCP/UDP and client/server awareness | ☐ RS-232/RS-485 and serial settings |
| ☐ Modbus RTU and Modbus TCP | ☐ Unit ID, function, register, offset, byte order, and data type |
| ☐ OPC UA server/client, tags, quality, and security awareness | ☐ Profinet, EtherNet/IP, timeouts, retries, and stale data awareness |

REQUIRED LEARNING OUTCOMES

- | | |
|---|--|
| ☐ Explain the module concepts without copying notes | ☐ Complete the calculations, diagrams, or logic correctly |
| ☐ Apply the content in the module practical activity | ☐ Diagnose the module case study systematically |
| ☐ Create the required engineering evidence | ☐ State safety boundaries and assumptions honestly |

COMPLETION GATE

- | | |
|---|---|
| ☐ Closed-book explanation completed | ☐ Exercises corrected and understood |
| ☐ Practical behavior matches written expectation | ☐ Fault root cause supported by evidence |
| ☐ Documents are readable, dated, and versioned | ☐ MODULE COMPLETE - date: _____ |

MODULE 11

Industrial networking and control-system communications

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Draw a PLC-HMI-SCADA-switch-VFD network	One-page concept artifact
2	Draw or calculate	Create an IP schedule and same-subnet checks	Calculation or annotated diagram
3	Apply in simulation	Build a ten-point Modbus map	Screenshot plus configuration/logic
4	Test and observe	Simulate Modbus TCP or OPC UA exchange	Expected-vs-actual result
5	Explain and document	Create a layered communication troubleshooting checklist	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|---|---|
| ☐ Check whether 192.168.10.20/24 and 192.168.11.30/24 share a subnet | ☐ Explain IP address versus register address |
| ☐ List causes when ping works but data does not | ☐ Explain register offset and byte order |
| ☐ Compare Modbus TCP and OPC UA | ☐ Write timeout and recovery tests |

CASE STUDY

SCADA can ping the PLC but tank level does not update

SITUATION	Pump status updates and ping succeeds, but tank level remains zero on SCADA.
AVAILABLE EVIDENCE	PLC level is 57.3 percent. The client reads register 40011. The mapping document assigns level to 40010. Other values work.
YOUR TASKS	☐ Identify working and failing layers ☐ Recognize the mapping issue ☐ Correct it in simulation ☐ Verify value, data type, scale, quality, and recovery
REQUIRED OUTPUT	Layered diagnosis, corrected protocol map, and communication test record.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 11

Industrial networking and control-system communications

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
11-17 Jan	Mon - 2 h	Cable/topology	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	PLC Ethernet settings	20 recall 50 learn 50 program/test
	Wed - 2 h	Smart instruments	20 recall 60 learn 40 apply
	Thu - 2 h	Data quality	20 recall 60 learn 40 process case
	Fri - 2 h	15 Jan - C: network docs	20 review 60 rotation 40 evidence
	Sat - 4 h	Build IP plan/link	60 plan 120 build 60 test/document
	Sun - 3 h	Link/IP fault	60 recall 60 fault 60 document/review
18-24 Jan	Mon - 2 h	Shielding/termination	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Modbus/OPC mapping	20 recall 50 learn 50 program/test
	Wed - 2 h	Serial/HART	20 recall 60 learn 40 apply
	Thu - 2 h	Timeout/fallback	20 recall 60 learn 40 process case
	Fri - 2 h	22 Jan - D: integration	20 review 60 rotation 40 evidence
	Sat - 4 h	Execute comms tests	60 plan 120 build 60 test/document
	Sun - 3 h	Register case; sign off	60 recall 60 fault 60 document/review

Project start

Saturday, 16 January 2027. Start after the relevant safety and readiness checks are complete.

Practical deliverable: Industrial communications laboratory

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|---|--|
| ☐ Network architecture | ☐ IP schedule |
| ☐ Protocol map | ☐ Successful exchange |
| ☐ Timeout test | ☐ Recovery test |
| ☐ Fault report | |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 12

SCADA integration, diagnostics, and historical data

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	25-31 January 2027	ROLE ALIGNMENT	Junior SCADA, control systems, PLC, and automation roles
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Module objective

Prove the complete path from process measurement to PLC logic, network, SCADA display, alarm, trend, command, and diagnostic evidence.

CORE CONTENT CHECKLIST

- | | |
|--|---|
| ☐ End-to-end tag lifecycle and naming | ☐ Command path versus feedback path |
| ☐ Read/write permissions and confirmation | ☐ Quality, timestamp, and stale data |
| ☐ Alarm event and acknowledgement path | ☐ Trend sampling, deadband, and history concepts |
| ☐ Communication watchdog | ☐ Store-and-forward and redundancy awareness |
| ☐ Backup, export, restore, and version concepts | ☐ Access control, least privilege, and integration testing |

REQUIRED LEARNING OUTCOMES

- | | |
|---|--|
| ☐ Explain the module concepts without copying notes | ☐ Complete the calculations, diagrams, or logic correctly |
| ☐ Apply the content in the module practical activity | ☐ Diagnose the module case study systematically |
| ☐ Create the required engineering evidence | ☐ State safety boundaries and assumptions honestly |

COMPLETION GATE

- | | |
|---|---|
| ☐ Closed-book explanation completed | ☐ Exercises corrected and understood |
| ☐ Practical behavior matches written expectation | ☐ Fault root cause supported by evidence |
| ☐ Documents are readable, dated, and versioned | ☐ MODULE COMPLETE - date: _____ |

MODULE 12

SCADA integration, diagnostics, and historical data

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Integrate at least 20 PLC tags	One-page concept artifact
2	Draw or calculate	Build a tag register with source, type, scale, access, quality, alarm, and display	Calculation or annotated diagram
3	Apply in simulation	Test five alarms end-to-end	Screenshot plus configuration/logic
4	Test and observe	Trend four related signals and explain an event	Expected-vs-actual result
5	Explain and document	Break communication, verify behavior, restore, and record recovery time	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|---|--|
| ☐ Trace one operator command to PLC and feedback | ☐ Explain connection versus correct mapping |
| ☐ Specify read-only and read/write tags | ☐ Compare scan, poll, and historian rates |
| ☐ Write communications acceptance criteria | ☐ Create a backup-and-restore checklist |

CASE STUDY

SCADA values update but operator commands are rejected

SITUATION	Status and process values update normally. Pressing Start has no effect and no PLC command tag changes.
AVAILABLE EVIDENCE	Connection is healthy. The Start tag is configured read-only. Local physical start works.
YOUR TASKS	☐ Separate network, permissions, access, PLC, and field possibilities ☐ Identify the cause ☐ Define confirmation and audit behavior ☐ Test start, stop, mode, and denied access
REQUIRED OUTPUT	Tag-access correction, command-path test, and cybersecurity-awareness note.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 12

SCADA integration, diagnostics, and historical data

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
25-31 Jan	Mon - 2 h	Control-panel review	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	PLC tag cleanup	20 recall 50 learn 50 program/test
	Wed - 2 h	Instrument quality	20 recall 60 learn 40 apply
	Thu - 2 h	Integrated process	20 recall 60 learn 40 process case
	Fri - 2 h	29 Jan - A: final screens	20 review 60 rotation 40 evidence
	Sat - 4 h	Integrate and FAT	60 plan 120 build 60 test/document
	Sun - 3 h	Command-access case	60 recall 60 fault 60 document/review

Project start

Saturday, 30 January 2027. Start after the relevant safety and readiness checks are complete.

Practical deliverable: SCADA Process Control System

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|--|--|
| ☐ Architecture | ☐ Tag register |
| ☐ Process overview | ☐ Alarm tests |
| ☐ Trends | ☐ Command tests |
| ☐ Comms-loss/recovery | ☐ Demo |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 13

Maintenance troubleshooting, testing, and change control

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	1-7 February 2027	ROLE ALIGNMENT	Critical employability module for all technician roles
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Module objective

Diagnose using evidence, preserve recoverability, test expected behavior, communicate results, and avoid uncontrolled changes.

CORE CONTENT CHECKLIST

☐ Symptom, root cause, and contributing condition	☐ Expected state and known-good reference
☐ Power-field-I/O-logic-output-actuator-process-HMI layers	☐ Fault isolation and divide-and-conquer
☐ Online monitoring, trends, alarms, and timestamps	☐ Forcing, bypass, and online-change risks
☐ Backup, version, checksum, rollback, and authorization	☐ FAT, SAT, loop check, I/O check, and functional-test concepts
☐ Expected and actual results	☐ Retest, regression, handover, escalation, and maintenance records

REQUIRED LEARNING OUTCOMES

☐ Explain the module concepts without copying notes	☐ Complete the calculations, diagrams, or logic correctly
☐ Apply the content in the module practical activity	☐ Diagnose the module case study systematically
☐ Create the required engineering evidence	☐ State safety boundaries and assumptions honestly

COMPLETION GATE

☐ Closed-book explanation completed	☐ Exercises corrected and understood
☐ Practical behavior matches written expectation	☐ Fault root cause supported by evidence
☐ Documents are readable, dated, and versioned	☐ MODULE COMPLETE - date: _____

MODULE 13

Maintenance troubleshooting, testing, and change control

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Build a pump expected-state table	One-page concept artifact
2	Draw or calculate	Create a check-basics-first fault flow	Calculation or annotated diagram
3	Apply in simulation	Inject and diagnose ten earlier-project faults	Screenshot plus configuration/logic
4	Test and observe	Build a controlled-change checklist	Expected-vs-actual result
5	Explain and document	Write an unresolved-fault handover without speculation	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|--|--|
| ☐ Separate symptom and cause in five examples | ☐ Choose first checks for pump-will-not-start |
| ☐ Explain how forcing can hide a fault | ☐ Write normal, fault, and recovery tests for a valve |
| ☐ Define when status stays NOT TESTED | ☐ Create rollback for a one-rung simulated change |

CASE STUDY Pump will not start after maintenance	
SITUATION	Auto is selected and a start is requested. HMI shows Start Permitted = False. PLC and network are healthy.
AVAILABLE EVIDENCE	E-stop, overload, and level permissives are healthy. Local/Remote feedback indicates Local; the selector was moved during maintenance.
YOUR TASKS	☐ Define expected state ☐ Identify the blocking permissive ☐ Explain why bypass is wrong ☐ Restore selector, verify operation, and document handover
REQUIRED OUTPUT	Troubleshooting report plus normal, fault, and recovery test records.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 13

Maintenance troubleshooting, testing, and change control

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
1-7 Feb	Mon - 2 h	Electrical test sequence	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	I/O/logic tracing	20 recall 50 learn 50 program/test
	Wed - 2 h	Loop isolation	20 recall 60 learn 40 apply
	Thu - 2 h	Process symptoms	20 recall 60 learn 40 process case
	Fri - 2 h	5 Feb - B: network diagnosis	20 review 60 rotation 40 evidence
	Sat - 4 h	Run fault pack	60 plan 120 build 60 test/document
	Sun - 3 h	Pump case; reports	60 recall 60 fault 60 document/review

Project start

Saturday, 6 February 2027. Start after the relevant safety and readiness checks are complete.

Practical deliverable: Fault-finding and test pack

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

- | | |
|--|--|
| ☐ Ten fault reports | ☐ Expected-state tables |
| ☐ PLC traces | ☐ HMI alarms |
| ☐ Trend evidence | ☐ Test records |
| ☐ Change log | ☐ Backup record |

VERIFICATION RECORD

- | | |
|---|--|
| ☐ Scope and assumptions written | ☐ Expected results written before testing |
| ☐ Normal behavior tested | ☐ At least two fault paths tested |
| ☐ Actual results recorded honestly | ☐ Unperformed tests marked NOT TESTED |
| ☐ Files backed up and versioned | ☐ Evidence explainable without notes |

MODULE 14

Flagship integrated automation system

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	8-28 February 2027	ROLE ALIGNMENT	Automation, controls, PLC, SCADA, and instrumentation trainee roles
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Module objective

Combine the junior competency range into one traceable simulation with controlled scope, reproducible tests, and honest limitations.

CORE CONTENT CHECKLIST

- | | |
|---|---|
| ☐ Requirements, scope, assumptions, exclusions, and acceptance | ☐ Raw Water to Pumping to Treatment Tank to Filtration to Treatment to Storage to Output |
| ☐ Equipment modules, modes, commands, status, and faults | ☐ Level, flow, pressure, temperature, 4-20 mA, and quality |
| ☐ Sequences, interlocks, permissives, trips, and recovery | ☐ PID only where justified |
| ☐ HMI, alarms, trends, diagnostics, and history | ☐ Ethernet, IP, Modbus TCP, OPC UA, and network-loss behavior |
| ☐ Complete engineering document set | ☐ FAT, defects, backups, versions, limitations, and demonstration |

REQUIRED LEARNING OUTCOMES

- | | |
|---|--|
| ☐ Explain the module concepts without copying notes | ☐ Complete the calculations, diagrams, or logic correctly |
| ☐ Apply the content in the module practical activity | ☐ Diagnose the module case study systematically |
| ☐ Create the required engineering evidence | ☐ State safety boundaries and assumptions honestly |

COMPLETION GATE

- | | |
|---|---|
| ☐ Closed-book explanation completed | ☐ Exercises corrected and understood |
| ☐ Practical behavior matches written expectation | ☐ Fault root cause supported by evidence |
| ☐ Documents are readable, dated, and versioned | ☐ MODULE COMPLETE - date: _____ |

MODULE 14

Flagship integrated automation system

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Freeze written scope before programming	One-page concept artifact
2	Draw or calculate	Build one process area at a time and test before integration	Calculation or annotated diagram
3	Apply in simulation	Audit every tag across documents and software	Screenshot plus configuration/logic
4	Test and observe	Execute normal, mode, fault, network-loss, and recovery FAT tests	Expected-vs-actual result
5	Explain and document	Record a ten-minute technical demonstration	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|--|---|
| ☐ Write five objective acceptance criteria | ☐ Create cause-and-effect for five critical events |
| ☐ Decide what remains local if SCADA fails | ☐ Trace one level tag across every layer |
| ☐ Define safe state for each pump and valve | ☐ Create defect register and retest process |

CASE STUDY

Network failure during treatment-tank filling

SITUATION	SCADA loses communication while the treatment tank fills automatically. PLC scan and local I/O remain healthy.
AVAILABLE EVIDENCE	High and high-high protection execute in the PLC. SCADA values become stale and remote commands are unavailable.
YOUR TASKS	☐ Define what continues locally ☐ Define HMI indication ☐ Decide whether the sequence continues, pauses, or stops ☐ Test protection, recovery, and ownership
REQUIRED OUTPUT	Failure philosophy, cause-and-effect update, event timeline, FAT results, and limitation statement.

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 14

Flagship integrated automation system

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
8-14 Feb	Mon - 2 h	Motor/valve interfaces	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Equipment modules	20 recall 50 learn 50 program/test
	Wed - 2 h	Instrument definitions	20 recall 60 learn 40 apply
	Thu - 2 h	Control narrative	20 recall 60 learn 40 process case
	Fri - 2 h	12 Feb - C: baseline docs	20 review 60 rotation 40 evidence
	Sat - 4 h	Design first subsystem	60 plan 120 build 60 test/document
	Sun - 3 h	Design review	60 recall 60 fault 60 document/review
15-21 Feb	Mon - 2 h	Protection audit	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Sequence/modes	20 recall 50 learn 50 program/test
	Wed - 2 h	PV validation	20 recall 60 learn 40 apply
	Thu - 2 h	Alarms/PID	20 recall 60 learn 40 process case
	Fri - 2 h	19 Feb - D: integration	20 review 60 rotation 40 evidence
	Sat - 4 h	Integrate PLC/HMI	60 plan 120 build 60 test/document
	Sun - 3 h	Network-failure case	60 recall 60 fault 60 document/review
22-28 Feb	Mon - 2 h	Schematic final	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Logic cleanup/backup	20 recall 50 learn 50 program/test
	Wed - 2 h	Tag audit	20 recall 60 learn 40 apply
	Thu - 2 h	Recovery behavior	20 recall 60 learn 40 process case
	Fri - 2 h	26 Feb - A: final HMI	20 review 60 rotation 40 evidence
	Sat - 4 h	Execute FAT/demo	60 plan 120 build 60 test/document
	Sun - 3 h	Close defects; sign off	60 recall 60 fault 60 document/review

Project start

Saturday, 13 February 2027. Start after the relevant safety and readiness checks are complete.

Practical deliverable: Automated Water Treatment Plant - Personal Training / Simulation Project

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

☐ Process/P&ID;	☐ Architecture
☐ I/O and instruments	☐ PLC logic
☐ HMI/SCADA	☐ Network plan
☐ FAT	☐ Fault reports
☐ Demo	☐ README

VERIFICATION RECORD

☐ Scope and assumptions written	☐ Expected results written before testing
☐ Normal behavior tested	☐ At least two fault paths tested
☐ Actual results recorded honestly	☐ Unperformed tests marked NOT TESTED
☐ Files backed up and versioned	☐ Evidence explainable without notes

MODULE 15

Portfolio, technical interviews, and targeted applications

LEARN | EXPLAIN | APPLY | TROUBLESHOOT | DOCUMENT

DATES	1-31 March 2027	ROLE ALIGNMENT	Employment readiness for all listed junior and trainee roles
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Module objective

Turn completed technical work into concise evidence and explain it honestly in applications, interviews, and demonstrations.

CORE CONTENT CHECKLIST

- | | |
|--|--|
| ☐ CV alignment and project bullets | ☐ Portfolio evidence register |
| ☐ GitHub structure and READMEs | ☐ Studied, simulated, implemented, tested, documented, and certified labels |
| ☐ Two-minute introduction | ☐ Ten-minute project walkthrough |
| ☐ Electrical, PLC, instrumentation, process, SCADA, network, and safety questions | ☐ Evidence-based troubleshooting explanation |
| ☐ Qualification fit, supervision, and limitations | ☐ Targeted applications, tracking, feedback, and continued improvement |

REQUIRED LEARNING OUTCOMES

- | | |
|---|--|
| ☐ Explain the module concepts without copying notes | ☐ Complete the calculations, diagrams, or logic correctly |
| ☐ Apply the content in the module practical activity | ☐ Diagnose the module case study systematically |
| ☐ Create the required engineering evidence | ☐ State safety boundaries and assumptions honestly |

COMPLETION GATE

- | | |
|---|---|
| ☐ Closed-book explanation completed | ☐ Exercises corrected and understood |
| ☐ Practical behavior matches written expectation | ☐ Fault root cause supported by evidence |
| ☐ Documents are readable, dated, and versioned | ☐ MODULE COMPLETE - date: _____ |

MODULE 15

Portfolio, technical interviews, and targeted applications

ACTIVITIES, EXERCISES, AND TECHNICIAN CASE STUDY

Complete the activity before reading or discussing the case-study answer. Predict expected behavior first and save evidence while you work.

REQUIRED ACTIVITIES

No.	Activity type	Task	Evidence
1	Build understanding	Audit every public claim	One-page concept artifact
2	Draw or calculate	Create a skill-to-evidence index	Calculation or annotated diagram
3	Apply in simulation	Record 90-second, five-minute, and ten-minute project explanations	Screenshot plus configuration/logic
4	Test and observe	Complete four mock interviews and create revision cards	Expected-vs-actual result
5	Explain and document	Map five real job descriptions to fit, gaps, and application decision	Short explanation or report

EXERCISES - ANSWER BEFORE CHECKING NOTES

- | | |
|---|--|
| ☐ Explain scan cycle, 4-20 mA, interlock, and permissive without notes | ☐ Diagnose pump-will-not-start aloud |
| ☐ Explain what simulation proves and does not prove | ☐ Write one CV bullet per major project |
| ☐ Prepare an honest no-industry-experience answer | ☐ Identify roles with formal qualification mismatch |

CASE STUDY		Recruiter challenges the value of simulation projects
SITUATION	An interviewer asks whether the water-treatment project was installed on a real plant and commissioned independently.	
AVAILABLE EVIDENCE	It is a personal simulation with PLC logic, HMI, documents, FAT results, and fault reports. It is not a live installation.	
YOUR TASKS	☐ Answer without exaggeration ☐ Explain what the evidence proves ☐ State what real-equipment learning remains ☐ Connect the work to supervised contribution	
REQUIRED OUTPUT	Rehearsed honest answer, concise demonstration, and evidence index.	

Case-study rule

Do not guess or change multiple things at once. State expected behavior, isolate the failing layer, choose a safe test, support the root cause with evidence, correct it, verify recovery, and record the result.

☐ CASE STUDY COMPLETE Result: _____ Date: _____ Revisit needed: YES / NO

MODULE 15

Portfolio, technical interviews, and targeted applications

DATED MONDAY-TO-SUNDAY EXECUTION PLAN

Week	Day	Module focus	Session method/output
1-7 Mar	Mon - 2 h	Electrical revision	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	PLC cleanup	20 recall 50 learn 50 program/test
	Wed - 2 h	Instrumentation revision	20 recall 60 learn 40 apply
	Thu - 2 h	Process/PID explanation	20 recall 60 learn 40 process case
	Fri - 2 h	5 Mar - B: network review	20 review 60 rotation 40 evidence
	Sat - 4 h	Portfolio audit	60 plan 120 build 60 test/document
	Sun - 3 h	Mock interview 1	60 recall 60 fault 60 document/review
8-14 Mar	Mon - 2 h	Schematics quiz	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	PLC fault drill	20 recall 50 learn 50 program/test
	Wed - 2 h	4-20 mA calculations	20 recall 60 learn 40 apply
	Thu - 2 h	SCADA/alarm review	20 recall 60 learn 40 process case
	Fri - 2 h	12 Mar - C: document audit	20 review 60 rotation 40 evidence
	Sat - 4 h	Record demo	60 plan 120 build 60 test/document
	Sun - 3 h	Mock interview 2; apply	60 recall 60 fault 60 document/review
15-21 Mar	Mon - 2 h	Motor/VFD review	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Logic-from-spec	20 recall 50 learn 50 program/test
	Wed - 2 h	Instrument fault	20 recall 60 learn 40 apply
	Thu - 2 h	Trend analysis	20 recall 60 learn 40 process case
	Fri - 2 h	19 Mar - D: integration	20 review 60 rotation 40 evidence
	Sat - 4 h	Tailor applications	60 plan 120 build 60 test/document
	Sun - 3 h	Mock interview 3	60 recall 60 fault 60 document/review
22-28 Mar	Mon - 2 h	Safety/isolation	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	PLC/HMI comms	20 recall 50 learn 50 program/test
	Wed - 2 h	Loop troubleshooting	20 recall 60 learn 40 apply
	Thu - 2 h	Narrative review	20 recall 60 learn 40 process case
	Fri - 2 h	26 Mar - A: HMI review	20 review 60 rotation 40 evidence
	Sat - 4 h	Submit applications	60 plan 120 build 60 test/document
	Sun - 3 h	Mock interview 4	60 recall 60 fault 60 document/review
29-31 Mar	Mon - 2 h	Final electrical recall	20 recall 60 learn 40 calculate/draw
	Tue - 2 h	Final PLC recall	20 recall 50 learn 50 program/test
	Wed - 2 h	Final instrument recall	20 recall 60 learn 40 apply
	Thu - 2 h	Continue applications	20 recall 60 learn 40 process case
	Fri - 2 h	Carry Friday cycle	20 review 60 rotation 40 evidence
	Sat - 4 h	Maintain portfolio	60 plan 120 build 60 test/document
	Sun - 3 h	Monthly review	60 recall 60 fault 60 document/review

MODULE 15

Portfolio, technical interviews, and targeted applications

APPLICATION PACK, PROJECT, AND VERIFICATION

March stage	Primary action	Required output
Week 1	Audit evidence and unsupported claims	Gap list and priority fixes
Week 2	Record demonstration and tailor applications	Demo plus targeted submissions
Week 3	Practise technical interview and logic-from-spec	Score sheet and corrections
Week 4	Submit focused applications and rehearse	Application tracker and interview narrative
Final days	Review responses and continue applying	Updated tracker and next plan

Project start

Saturday, 6 March 2027. Start after the relevant safety and readiness checks are complete.

Practical deliverable: Application readiness pack and technical demonstration

Use this deliverable to combine the module content. Record expected results before testing and keep unperformed items marked NOT TESTED.

EVIDENCE TO SAVE

☐ Final CV	☐ Portfolio
☐ GitHub READMEs	☐ Evidence index
☐ Mock interview scores	☐ Demo recordings
☐ Application tracker	☐ Gap log

VERIFICATION RECORD

☐ Scope and assumptions written	☐ Expected results written before testing
☐ Normal behavior tested	☐ At least two fault paths tested
☐ Actual results recorded honestly	☐ Unperformed tests marked NOT TESTED
☐ Files backed up and versioned	☐ Evidence explainable without notes

Competency and portfolio release standard

Level	Meaning	Evidence
1 - Recognition	Identify concept, device, symbol, or term	Correct identification
2 - Explanation	Explain what, why, and how in your own words	Closed-book answer
3 - Application	Use it in calculation, logic, diagram, configuration, or test	Working artifact
4 - Troubleshooting	Diagnose, correct, and verify a realistic fault	Evidence-based fault report
5 - Teaching	Explain operation and limits clearly to another person	Demonstration or teach-back

TARGET BY MARCH 2027

- | | |
|---|--|
| ☐ Level 3/4: electrical calculations, schematics, PLC scan and I/O | ☐ Level 3/4: Ladder Logic, modes, interlocks, permissives, and faults |
| ☐ Level 3/4: 4-20 mA scaling and loop fault reasoning | ☐ Level 3/4: systematic troubleshooting and verification |
| ☐ Level 3: HMI/SCADA, alarms, trends, and communication quality | ☐ Level 3: industrial networking and protocol diagnostics |
| ☐ Level 3: P&ID; I/O, narrative, alarm, network, test, and fault documents | ☐ Level 2/3: VFD, PID, HART, Profinet, and EtherNet/IP awareness |

Portfolio release gate

Every screenshot must be explainable. Every test must show expected and actual results. Every simulation must be labelled. Unperformed work remains NOT TESTED. No employment, certification, commissioning, or field experience may be implied without evidence.

Technician troubleshooting and case-study method

Stage	Required behavior
1. Make safe	Identify hazards, state, stored energy, and authorization limits.
2. Define	Write expected behavior, actual behavior, timing, and recent changes.
3. Check basics	Power, safety state, mode, permissives, device status, links, and alarms.
4. Trace	Follow field device -> I/O -> tag -> logic -> output -> actuator -> process -> HMI.
5. Isolate	Use evidence and known-good comparison; avoid random changes.
6. Test	Choose a safe test and write its expected result before performing it.
7. Diagnose	Name the root cause and separate it from symptoms.
8. Correct	Make one authorized, controlled correction with rollback where relevant.
9. Verify	Repeat failed step, normal test, fault test, and recovery test.
10. Document	Record evidence, cause, action, result, version, and remaining risk.

FINAL EVIDENCE AUDIT

- | | |
|--|--|
| ☐ At least ten complete troubleshooting reports | ☐ Normal, abnormal, and recovery tests |
| ☐ Readable PLC and HMI screenshots | ☐ P&ID, I/O, instrument, narrative, and alarm documents |
| ☐ Network and protocol maps | ☐ Backups, versions, change log, and limitations |
| ☐ Two-minute introduction | ☐ Ten-minute flagship demonstration |

Role-alignment references and scope note

Role patterns were checked against recent vacancy descriptions. Requirements vary by employer, country, industry, and legal framework; each job advert remains the authority.

Reference	Why it informed this syllabus	URL
Adsyst - Graduate / Junior Control Systems Technician	PLC, SCADA, HMI, VSD, instrumentation, design, testing, and project lifecycle	https://careers.adsyst.co.uk/job/886184
Amcor - Controls Technician	Electrical/PLC/HMI faults, drawings, networks, VFDs, sensors, safety, and records	https://amcor.wd5.myworkdayjobs.com/en-US/Amcor_External_Career_Site/job/Controls-Technician_REQ_90979
South African public procurement examples	Shows that technician roles may specify electrical, instrumentation, trade, software, or OEM qualification routes	https://www.etenders.gov.za/

Scope note

This is a personal study workbook, not a formal qualification, apprenticeship, trade test, site authorization, or safety certificate. Its purpose is to build explainable junior-level knowledge and evidence for supervised entry-level work.